

Overview of Data Science

Lecture-1

Introduction

The simplest definition of data science is the extraction of actionable insights from raw data.

For example, if we collect data about six students

As -

Student	Height
1	5 ft
2	6 ft
3	5.5 ft
4	5.8 ft
5	5.3 ft
6	4.8 ft

We can extract the following information -

- The average height of the student = 5.4 ft.
- The maximum height among the students is 6 ft
- The minimum height among the students is 4.8 ft.

Probably we can extract more information as -
The difference of heights of the students with respect to the mean height is -

Student	difference of height	Average difference
1	0.4 ft	0.44 ft
2	0.6 ft	
3	0.5 ft	
4	0.4 ft	
5	0.1 ft	
6	0.6 ft	

This example ~~is~~ illuminates the purpose of data science.

Therefore we can summarize the objectives of data science in the following way —

- i) Data science is a deep study of the massive amount of data to extract meaningful insights from raw, structured and unstructured data.
- ii) ~~Ext~~ To extract the meaningful and actionable data information from the data we use different scientific methods, different technologies and algorithms.
- iii) Data science is a multidisciplinary field that uses tools and techniques to manipulate the data so that ^{we} you can find something new and meaningful.
- iv) Data science uses the most powerful hardware, programming systems, and most efficient algorithms to solve the data related problems. That is why data science is one of the important areas of Artificial intelligence.
- v) Data science is an inter-disciplinary field.

Before entering into the details of methodology and tools of data science let's quickly review the basic terminologies used in data science.

The volume or the masses of data —

<u>Unit</u>	<u>Value</u>	<u>Size</u>
b (bit)	0 or 1	1/8 of a byte.
B (byte)	8 bits	1 byte
KB (kilobyte)	1,000 bytes	1,000 bytes
MB (megabyte)	1×10^6 bytes	1,000,000 bytes.
GB (gigabyte)	1×10^9 bytes	1,000,000,000 bytes.
TB (terabyte)	1×10^{12} bytes	1,000,000,000,000 bytes.
PB (petabyte)	1×10^{15} bytes	1,000,000,000,000,000 bytes.
EB (exabyte)	1×10^{18} bytes	1,000,000,000,000,000,000 bytes.
ZB (zetta byte)	1×10^{21} bytes	1,000,000,000,000,000,000,000 bytes.
YB (yotta byte)	1×10^{24} bytes	1,000,000,000,000,000,000,000,000 bytes.

Very interestingly a ~~survey~~ survey report conducted by Google (Erich Schmidt (2010) Executive Chairman, Google found —

From the dawn of civilization until 2003, human humankind generated five exabyte of data (5×10^{18} bytes). But now we produce five exabyte of data every two days and the pace is accelerating. The reason behind it the massive improvement of technology.

Let us take a few examples —

- i) The internet has reaches over 5 billion customers.
- ii) Over 90% of people access the internet via mobile devices.

(iii) Amount of data consumed per year on an average is 79 ZB.

Face book
users share 240k photos every one minute ^{of the} ~~per~~ day.

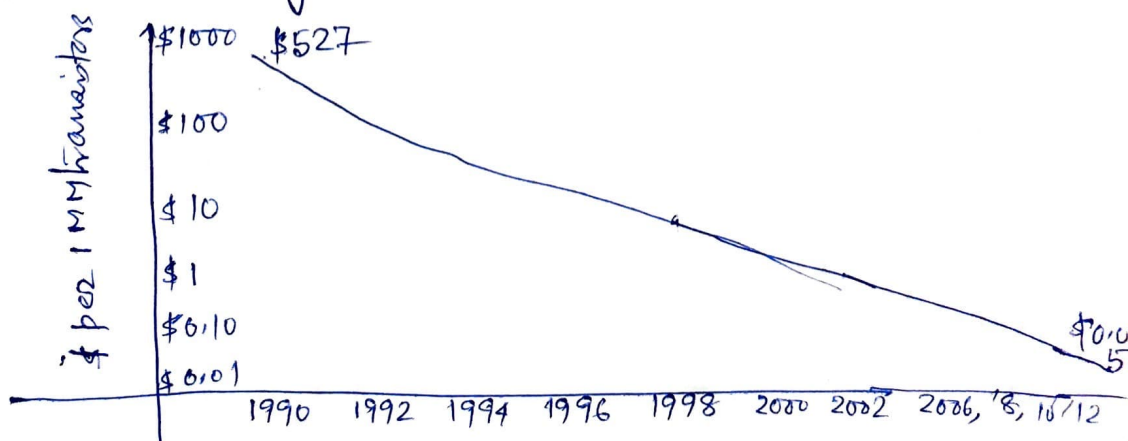
Face book live
44 M views every one minute ^{of the} ~~per~~ day.

Twitter
575 k tweets every one minute of the day

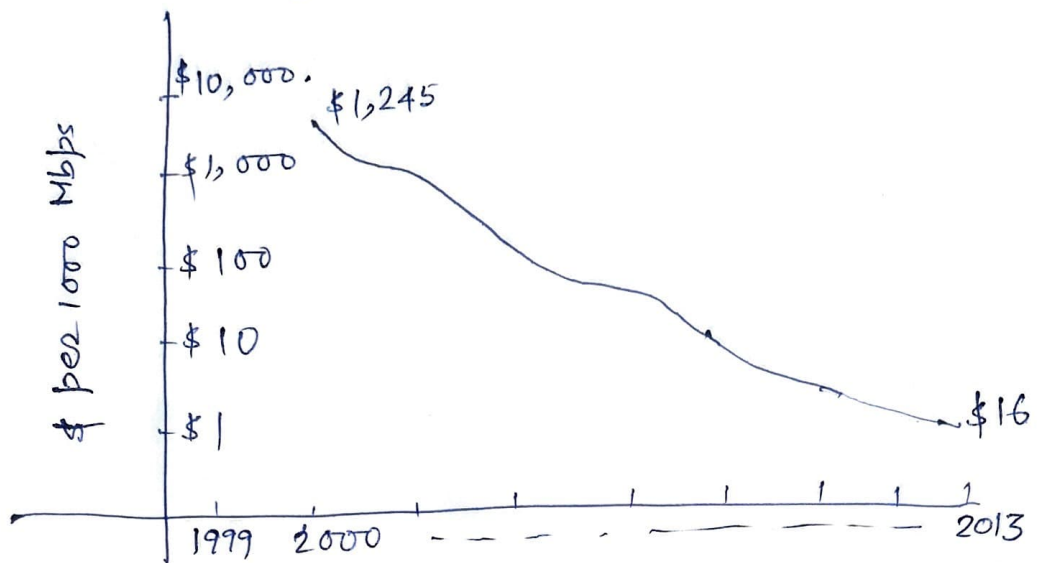
You Tube
894 k Hours of usage every one minute of the day.

One of the major reason is that the computer cost is declining rapidly due to the invention of new and improved technologies —

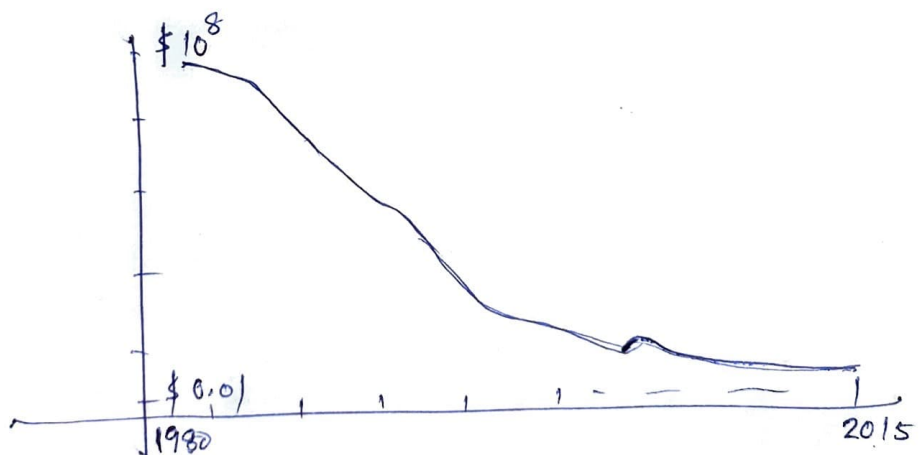
On an average the computing cost is declining by 33% Annually.



The second important issue is that the Bandwidth Costs is declining at a rate of 27% Annually.



The third major reason is the dropping of storage costs.

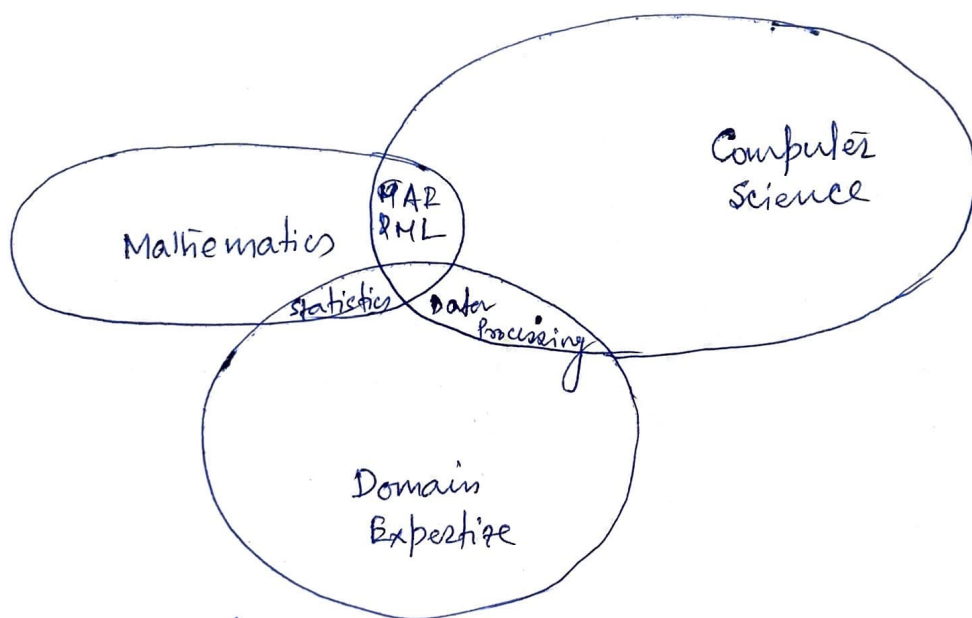


As a result of these factors we receive huge data and ~~to~~ it is termed as big data. It is not only the volume of data which is important rather there are other attributes which makes a set of data as big data.

Five V's of big-data

1. Volume — It defines the scale of data.
2. Velocity — Analysis of streaming data.
3. Variety — Different forms of data.
4. Veracity — Uncertainty of data.

To analyze such huge data we use the following knowledge resources.



Different aspects Usage of Data Science

- We can say that data science is all about:
- i) Asking the correct questions and analyzing the raw data.
 - ii) Modelling the data using various complex and efficient algorithms.
 - iii) Visualizing the data to get a better perspective.

- iv) understanding the data to make better decisions and finding the results.

Components of Data Science

- i) statistics
- ii) Domain Expertise,
- iii) Data Engineering,
- iv) Visualization,
- v) Advanced Computing.

Statistics: Statistics is one of the most important components of data science. It guides us to collect and analyze the numerical data in a large amount and finding insight from it.

Domain Experts: Domain expertise means specialized knowledge or skills of a particular area. In data science, there are various areas for which we need domain expertise.

Data Engineering: Data engineering is a part of data science, which involves acquiring, storing, retrieving and transforming the data. Data engineering also includes meta-data (data about data) to the data.

Visualization:

Data Visualization is meant by representing data in a visual context so that people can easily understand the significance of data. Data visualization makes it easy to access the huge amount of data in visuals.

Advanced Computing

Advanced computing involves designing, writing, debugging and maintaining the source code of computer programs.

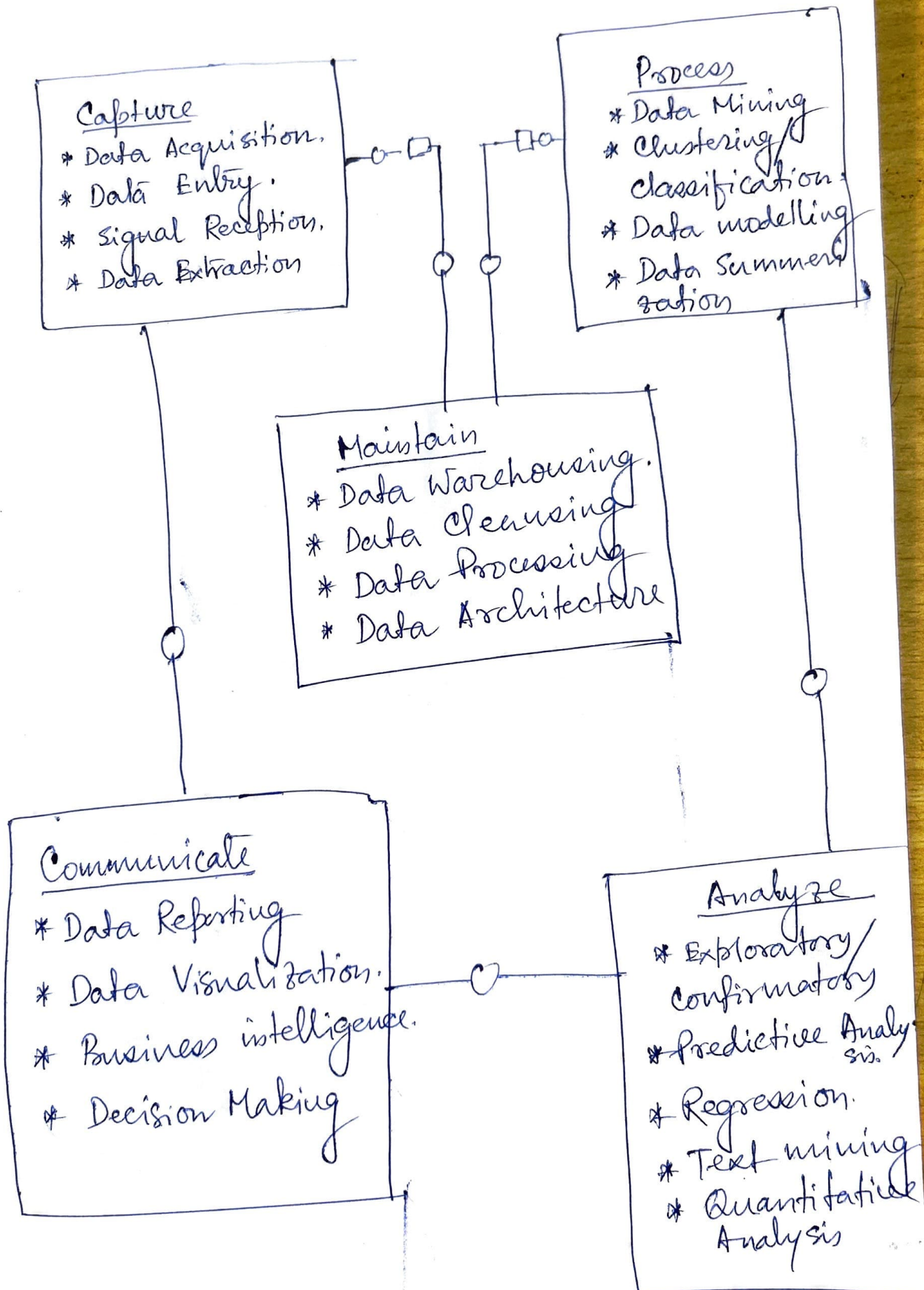
Mathematics

Mathematics involves the study of quantity, structure, space and changes. For a data scientist, knowledge of good mathematics is essential.

Machine Learning

Machine learning is all about to provide training to a machine so that it can act as a human brain.

Data Science Life Cycle



Tools Requiring For Data Science

Data Analysis tools: R, Python, Statistics, SAS, Jupyter, R-studio, MATLAB, Excel, Rapid Miner.

Data Warehousing: ETL, SQL, Hadoop, Informatica/Talend, AWS Redshift.

Data Visualization: R, Jupyter, Tableau, Cognos.

Machine Learning Tools: Spark, Mahout, Azure, ML Studio.

Applications of Data Science

1. Fraud and Risk Detection.
2. Healthcare
3. Internet Search.
4. Targeted Advertising.
5. Website Recommendations.
6. Image Recognition.
7. Speech Recognition.
8. Airline Route Planning.
9. Gaming
10. Augmented Reality.